

Fig. 3. GeoTex Bake10 lowest-absolute-error example at held-out camera 004. The Stone object has the lowest existing eleven-view FG-LPIPS among seven display-cleared assets, but Late-high minus Fixed Mean is only −0.000464 FG-LPIPS (+0.001 dB FG-PSNR). GT-256 reference self-bake, Fixed Mean, and Late-high use one common crop and the same mesh/UVs. This outcome-selected example separates absolute error from schedule benefit; it does not show a general effect or faithful material reproduction. The complete ten-object effect distribution is in Supplementary Section S5.

changed checkpoint and protocol make this a sensitivity boundary, not an independent backbone confirmation (Supplementary Section S8).

Native-pipeline quality boundary. On the same fifty inputs, the new geometry checkpoint trails full v29 by 2.16 dB FG-PSNR and has FG-LPIPS 0.0341 higher. This matched-input deployment comparison changes base weights and geometry implementation and uses a nondefault 50-step protocol, so it does not isolate TCAS. A same-base follow-up remains 0.0159 LPIPS worse than full v29 despite a local unscaled-versus-disabled geometry benefit. Neither result establishes TCAS deployment superiority; full tests and protocol details are in Supplementary Section S8.

4.5. Sensitivity Across Native Conditioning Interfaces

The auxiliary studies ask related scheduling questions under different native conditioning operators and camera contracts. The 50-object and GSO100 studies both use MV-Adapter; GSO100 is a separate source-ordered cohort and protocol sensitivity, not a third independent backbone. MVDiffusion scales a partial correspondence operator and is reported only as additional context. Table 7 presents each study's own specified primary contrast: the GeoTex contrast compares Middle-high with the mean of Early-high and Late-high, whereas both MV-Adapter contrasts compare Middle-high with Fixed Mean. We report these rows separately and do not pool them or treat them as an identical replication.

Input-selected native MV-Adapter study. We selected fifty Fresh C inputs not previously evaluated through MV-Adapter, using input availability and native material/geometry compatibility. Native mesh normalization and six orthographic cameras define both generation and evaluation targets. Three-object identity checks confirmed agreement between the deployed and recorded default paths. An earlier seven-object panel had an invalid camera calibration and is excluded from these comparisons (Supplementary Section S3).

Table 6. Native-camera MV-Adapter means on fifty input-selected objects.

Condition	FG-PSNR	FG-LPIPS	FG-SSIM	Edge-SSIM
Native / Fixed High	12.60	0.1280	0.5041	0.3526
Fixed Low	12.73	0.1270	0.5066	0.3475
Fixed Mean	12.68	0.1279	0.5068	0.3509
Early-high	12.60	0.1296	0.5041	0.3503
Middle-high	12.71	0.1262	0.5061	0.3470
Late-high	12.74	0.1260	0.5075	0.3514
Generic Linear	12.72	0.1263	0.5072	0.3501

The seven conditions share object inputs, seed, geometry and evaluation code. With 50 steps, low/high are 0.75/1.0; stage-high schedules contain 16 high steps, Fixed Mean is 0.83, and Generic Linear is 0.75–0.91. These five controls have matched per-feature sums within this interface. All 50 objects remain in the primary cohort, including thirteen texture-absent sources with legal constant-white base-color targets. Targets are unlit native-camera base color, not Fresh C's differently normalized reference renders or certified physical albedo.

The internally predeclared primary, Middle-high minus Fixed Mean FG-PSNR, is +0.029 dB (95% object-bootstrap CI [−0.019, +0.081]; paired Monte Carlo sign-flip $p = 0.281$; 29/50 favorable). This does not support a stable middle-window gain or establish equivalence. The remaining 104 pair–metric tests use one Holm family. Middle-high has lower Edge-SSIM than Fixed Mean (adjusted $p = 0.00104$); some other placement contrasts separate, including Late-high versus Early-high PSNR. Such endpoint-dependent differences do not demonstrate transferable C3 optimality. Complete contrasts, correction rules and the textured subset are in Supplementary Section S3. This is a native-interface validation of the scheduling question, not architecture-matched replication of GeoTex's capped residual intervention.

Separate external-input MV-Adapter sensitivity. The source-ordered GSO100 convenience cohort uses the same native MV-Adapter control path and three shared seeds. It includes seven scale/time profiles, with four inherited profiles frozen after ear-



Fig. 3. GeoTex Bake10 appearance at held-out camera 004. Each row shows a GT-256 reference self-bake, Fixed Mean, and Late-high under a shared mesh/UV and within-object crop. Tire is selected by median source-foreground Laplacian variance among seven display-cleared assets. The carved font is an explicitly post-outcome example with the largest existing eleven-view Late-high minus Fixed Mean FG-LPIPS reduction among those seven (−0.00781); its FG-PSNR also decreases by 0.271 dB. The selected examples illustrate appearance mismatch and an endpoint tradeoff, not cohort prevalence or a general schedule benefit. Source images are 512×512 and displayed without method-specific color edits. The full ten-object effect distribution and scores are in Supplementary Section S5.

nonreference PSNR and +0.00266 nonreference LPIPS; C3 has no corrected advantage over Fixed Mean or Fixed High. The changed checkpoint and protocol make this a sensitivity boundary, not an independent backbone confirmation (Supplementary Section S8).

Native-pipeline quality boundary. On the same fifty inputs, the new geometry checkpoint trails full v29 by 2.16 dB FG-PSNR and has FG-LPIPS 0.0341 higher. This matched-input deployment comparison changes base weights and geometry implementation and uses a nondefault 50-step protocol, so it does not isolate TCAS. A same-base follow-up remains 0.0159 LPIPS worse than full v29 despite a local unscaled-versus-disabled geometry benefit. Neither result establishes TCAS deployment superiority; full tests and protocol details are in Supplementary Section S8.

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